Evolutionary multiobjective optimization

Andrzej Jaszkiewicz

The problem of multiobjective optimization

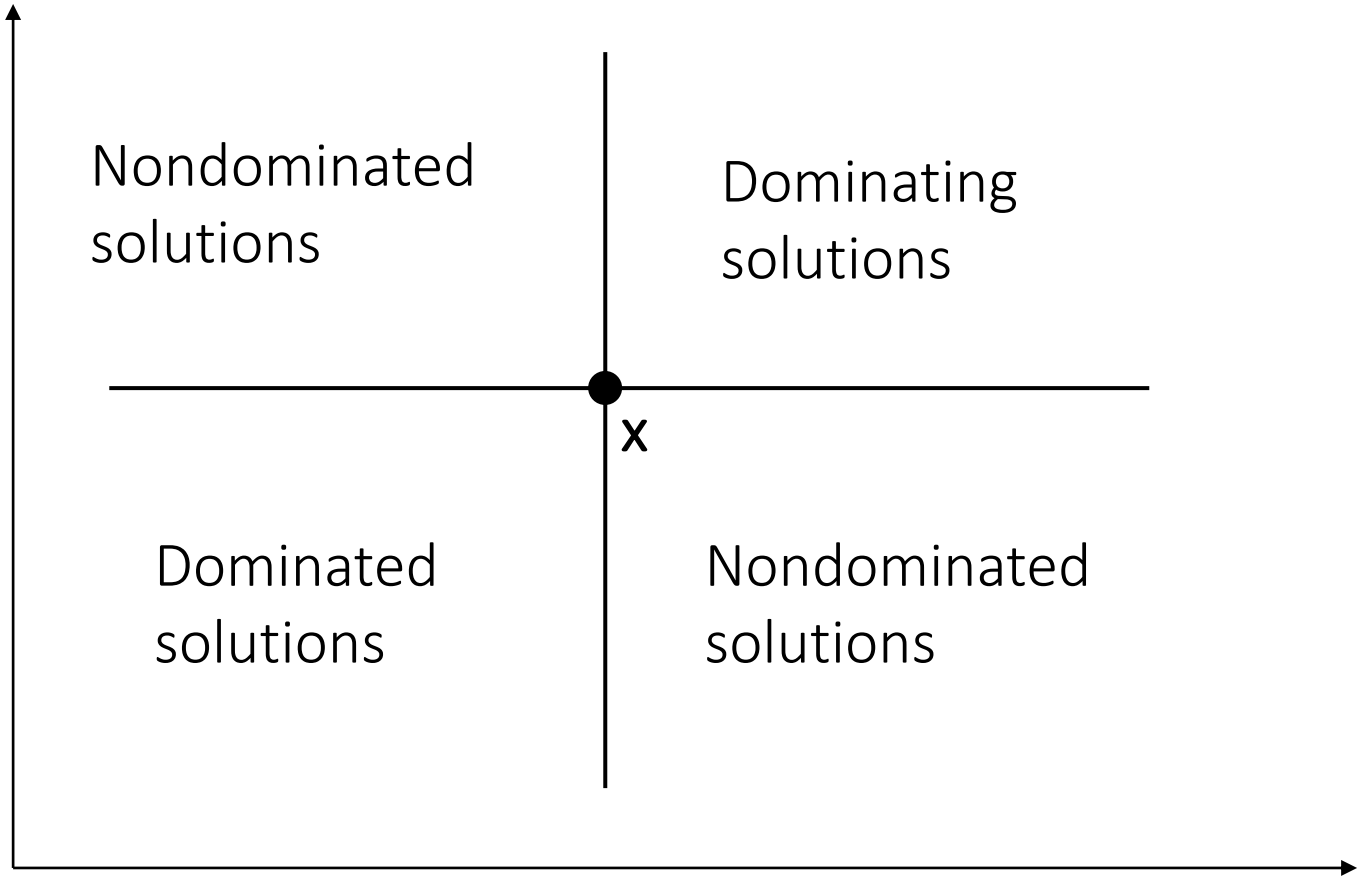
Maximize/Minimize $z_1 = f_1(\mathbf{x})$

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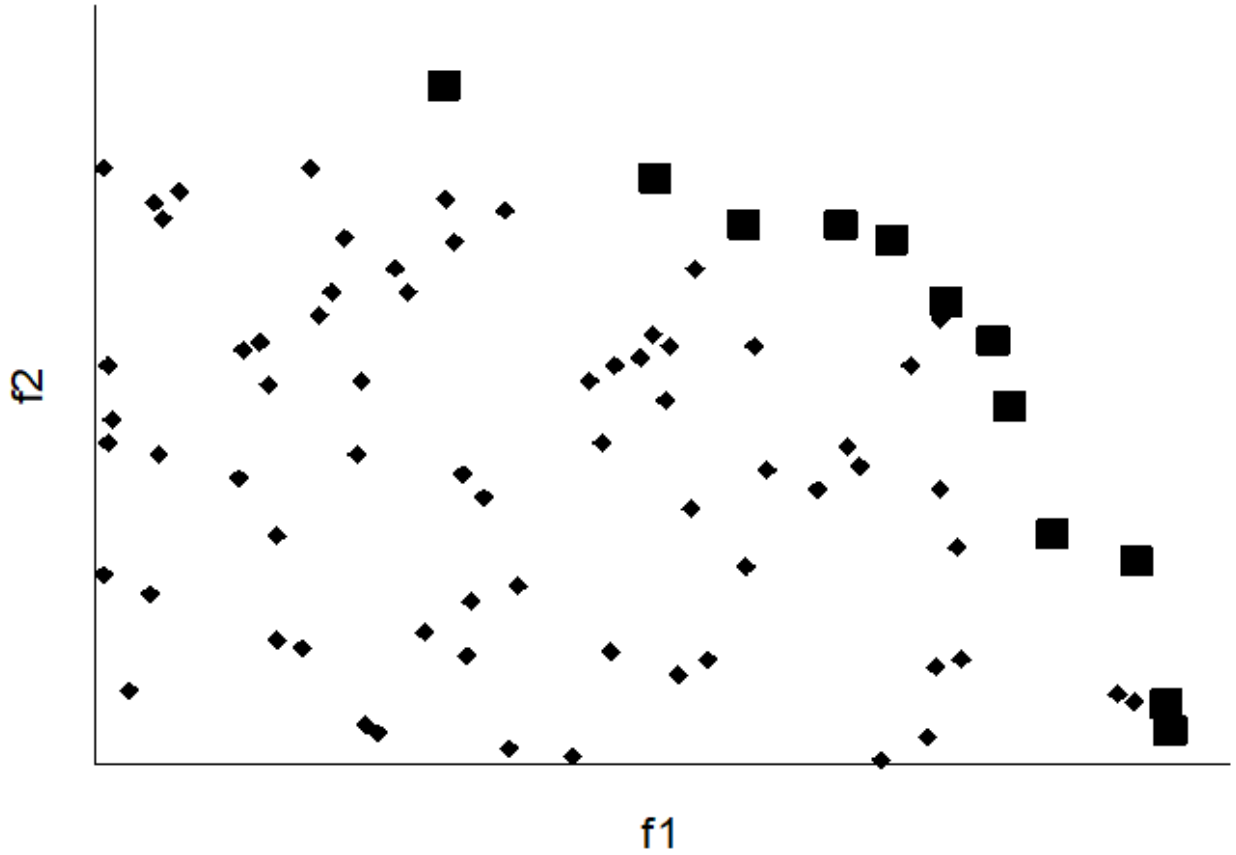
Maximize/Minimize $z_j = f_j(\mathbf{x})$

s.t. $\mathbf{x} \in D$

Dominance relation (maximization)



Pareto-optimal solutions – image in the objective space



Why Pareto-optimal solutions?

- The final goal: choosing a single solution
- Under a generally accepted assumption:

  The decision-maker always prefers to improve one objective without worsening the others – the compatibility of preferences with the dominance relation

  ⇒ the best solution belongs to the set of Pareto-optimal solutions

Classification of Methods for Solving Multiobjective Optimization Problems

- **A priori** preference specification
  - E.g. building a utility function and then optimizing it
- **Interactive** preference specification
  - E.g. successive scalarizing functions in subsequent iterations
- **A posteriori** preference specification
  - First, the generation of (approximately) Pareto-optimal solutions, then the selection

Single objective EAs/MHs in multiobjective optimization

- **A priori** preference specification
  - E.g. optimization of the utility functions
- **Interactive** preference specification
  - E.g. optimization of the scalarizing functions
- **A posteriori** preference specification
  - E.g. a series of optimization problems for different scalarizing functions solved by an EC/MH

The goal of multiobjective metaheuristics

- Generate in a single run a set of solutions that are an approximation of the whole or part of the set of Pareto-optimal solutions (a posteriori or interactive approach)
- Approximation:
  - Not all Pareto-optimal solutions need to be found
  - Approximate solutions may be dominated, but should be close to Pareto-optimal

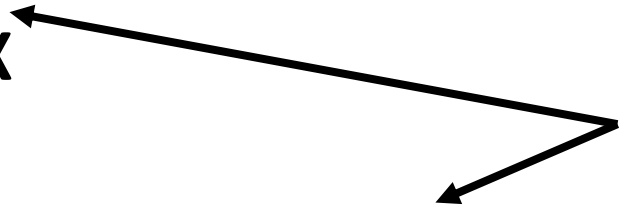
Selection pressure in a multiobjective case

Initialize memory **M**

Generate and evaluate the initial population **X**

Update **M** based on **X**

repeat



How to evaluate solutions with multiple objectives

Based on population **X**, its evaluation, and **M** generate and evaluate population **X'**

Update **M** based on **X’**

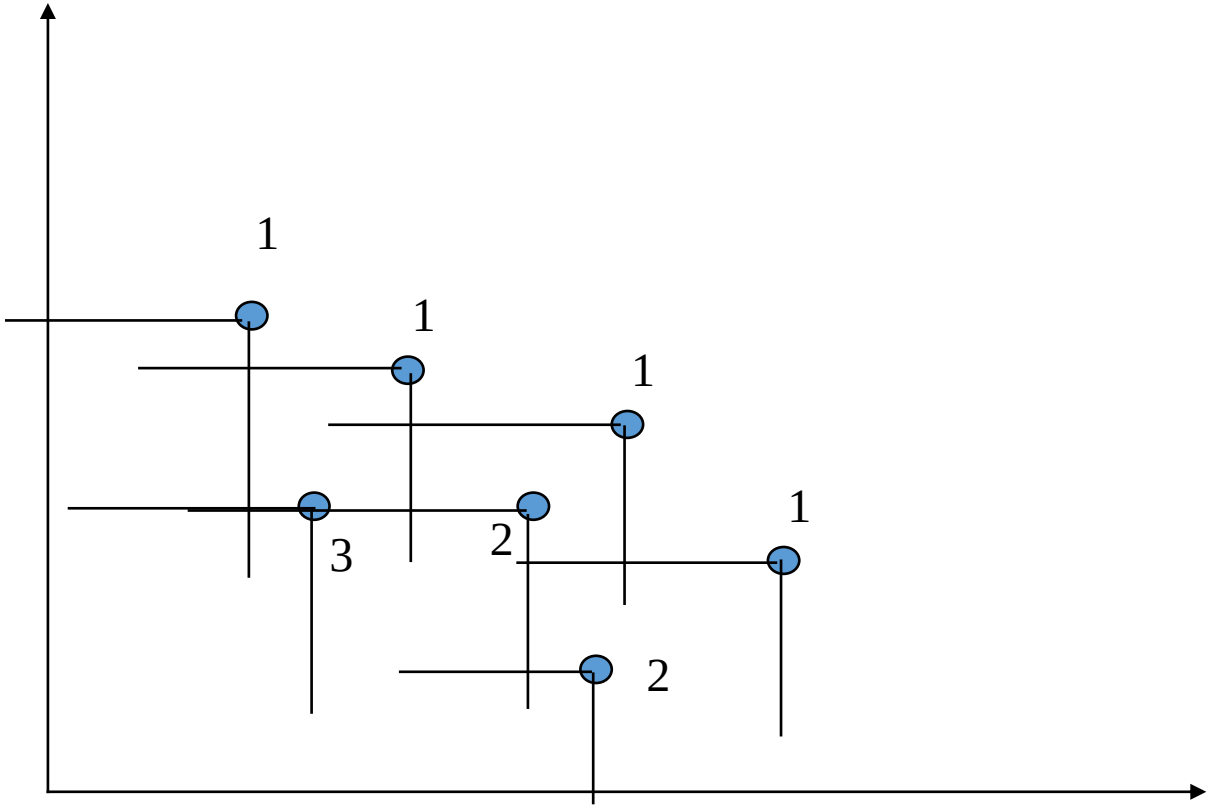
From **X**∪ **X’** select a new population **X**

until stopping conditions are met

Classification of methods

- Methods based on the dominance relation
  - Pareto-ranking based evolutionary algorithms – NSGA-II
  - Pareto local search
- Decomposition/scalarization-based algorithms
  - MOEA/D, MOGLS
- Indicator-based algorithms
  - HypE, SMS-EMOA, FV-MOEA

Pareto Ranking



Characteristics of selection pressure based on a dominance relation

+ Independence from objectives scaling (even non-linear)

+ No or few parameters

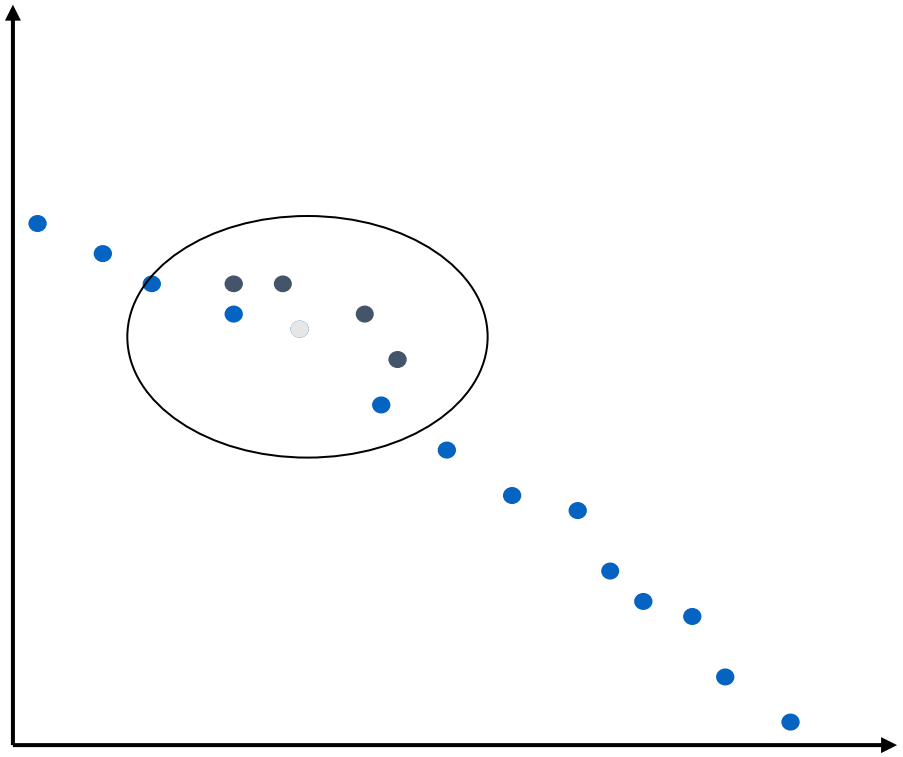
- All (localy) Pareto-optimal solutions have the same rank – no selection pressure to disperse/evenly distribute solutions

- Many local moves may not change the ranking

- Rankings become "flat" with an increase in the number of objectives

- Evaluation may be relatively expensive

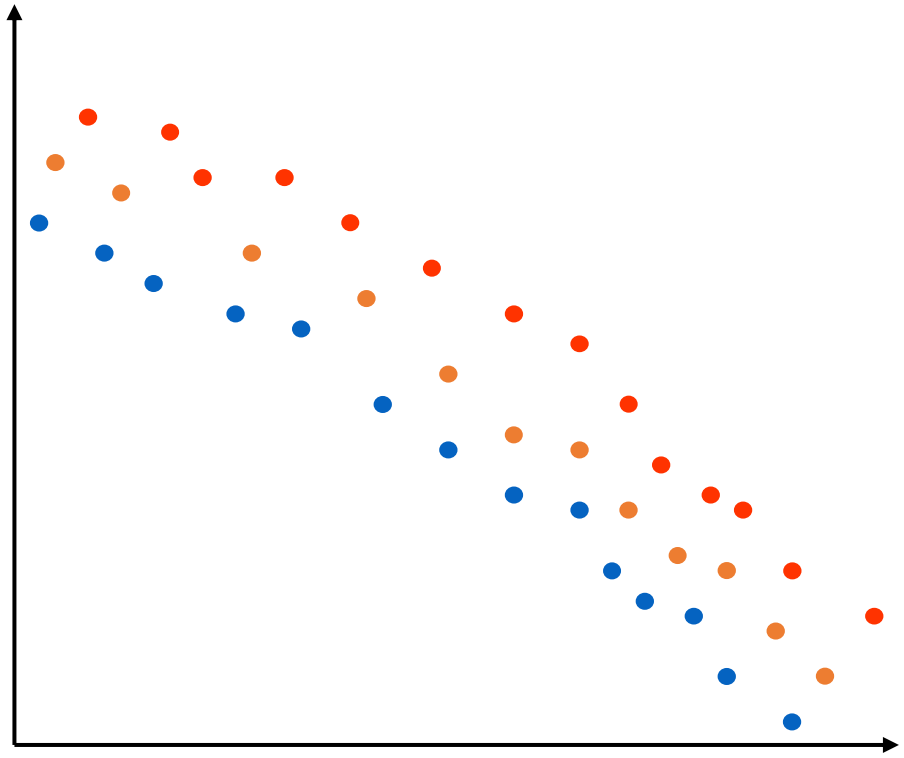
Pareto Local Search



Pareto Local Search

In each iteration

- For each potentially Pareto-optimal (PPO) solution x
  - Find all locally non-dominated solutions in N(x) and update the PPO set



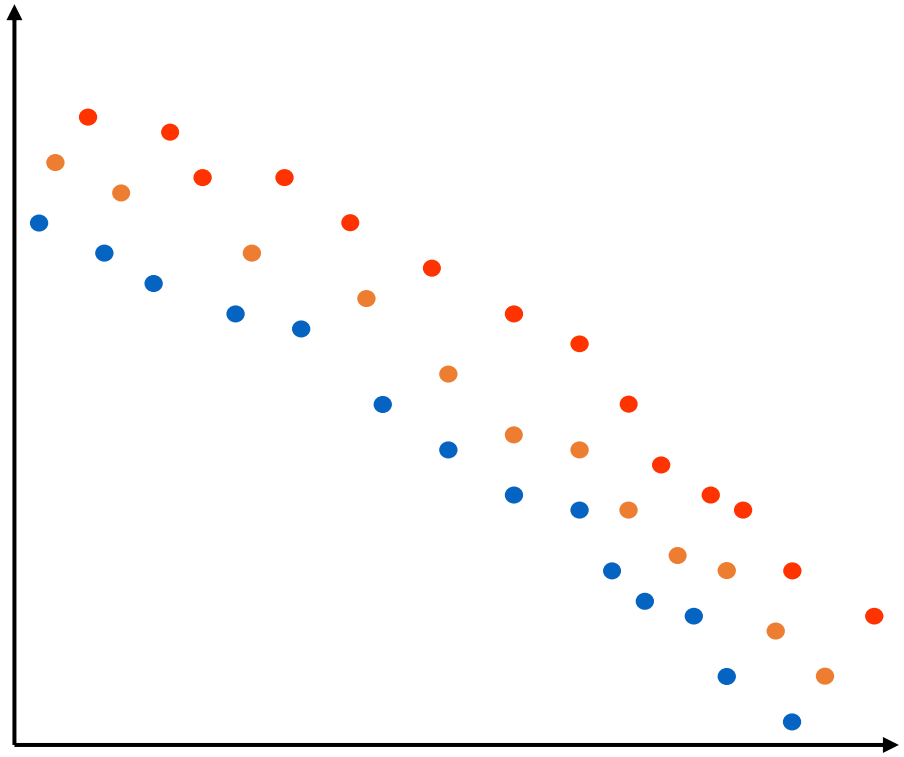
A different point of view – local search on a PPO set

Generate initial PPO set

powtarzaj

Browse the neighborhood of the current PPO and update the PPO

until the PPO was modified

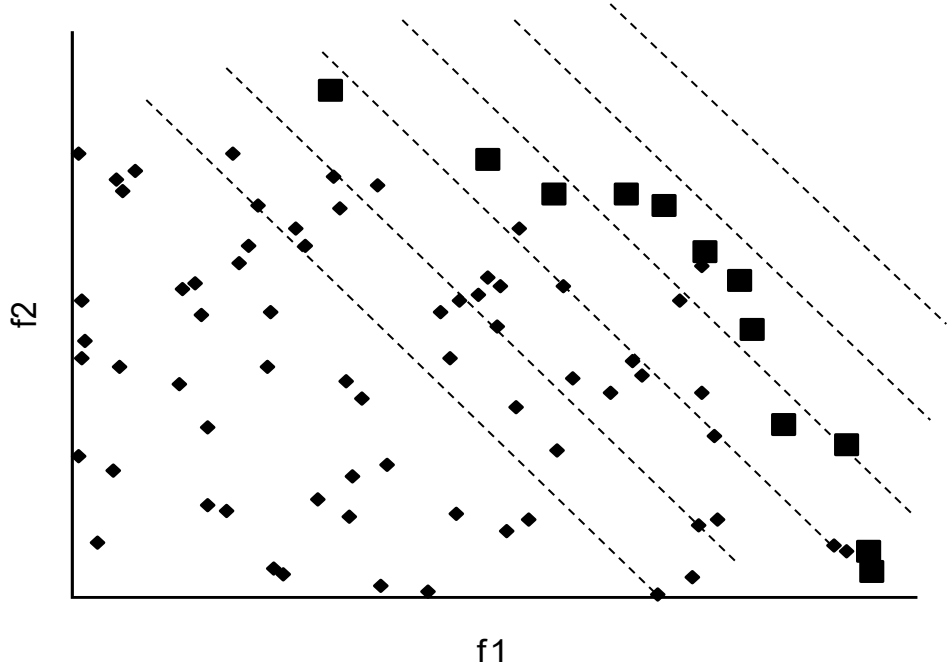


Pareto Local Search - characteristics

- Very inefficient when starting from poor solutions
- Very effective (best known results for many bi-objective problems) when starting from a sample of solutions close to Pareto-optimal
- Rapid decline in efficiency for three or more criteria
  - Local Pareto search for more than two criteria – selection of promising solutions for neighborhood exploration, use of ND-Tree
    - Many-objective Pareto local search, A Jaszkiewicz, European Journal of Operational Research 271 (3), 1001-1013, 2018

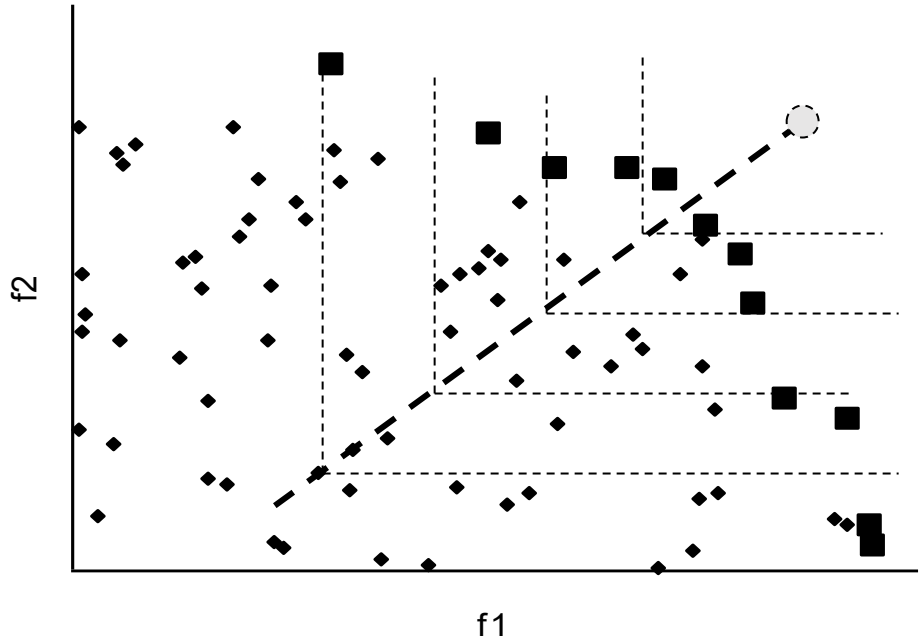
Linear scalarizing function

- $s_1(\mathbf{z}, \Lambda) = \sum_{j=1}^{J} \lambda_j z_j$
- Each linear scalarizing function has at least one optimum in the set of Pareto-optimal solutions



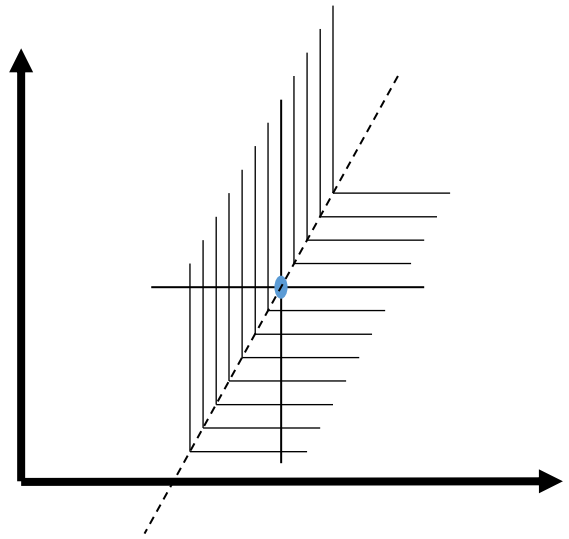
Chebyshev scalarizing function

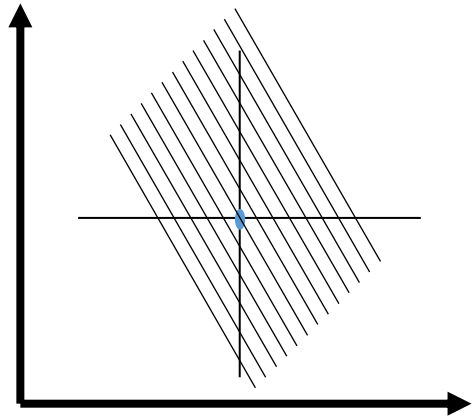
- $s_\infty(\mathbf{z}, \mathbf{z}^0, \Lambda) = \max_j\{\lambda_j(z_j^0 - z_j)\}$
- For every Pareto-optimal solution, there exists a Chebyshev scalarizing function for which this solution is the optimal
- Every Chebyshev scalarizing function h set of Pareto-optimal solutions



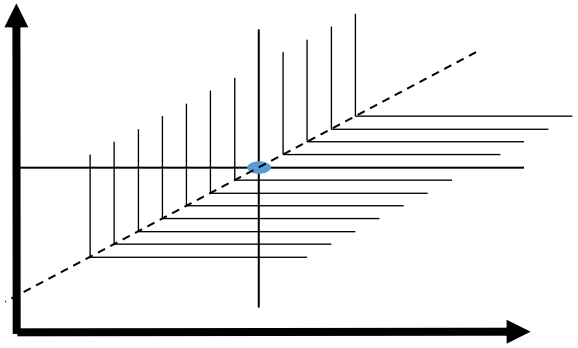
The role of weights in scalarizing functions

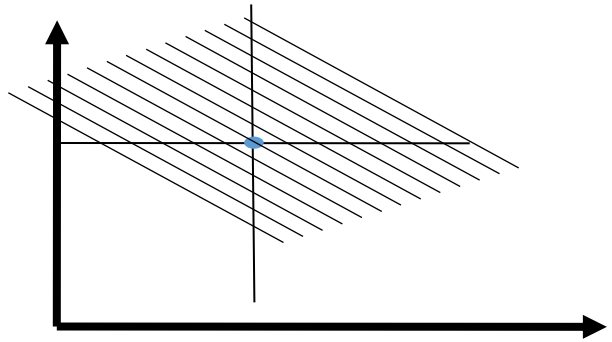
$\lambda_1 > \lambda_2$



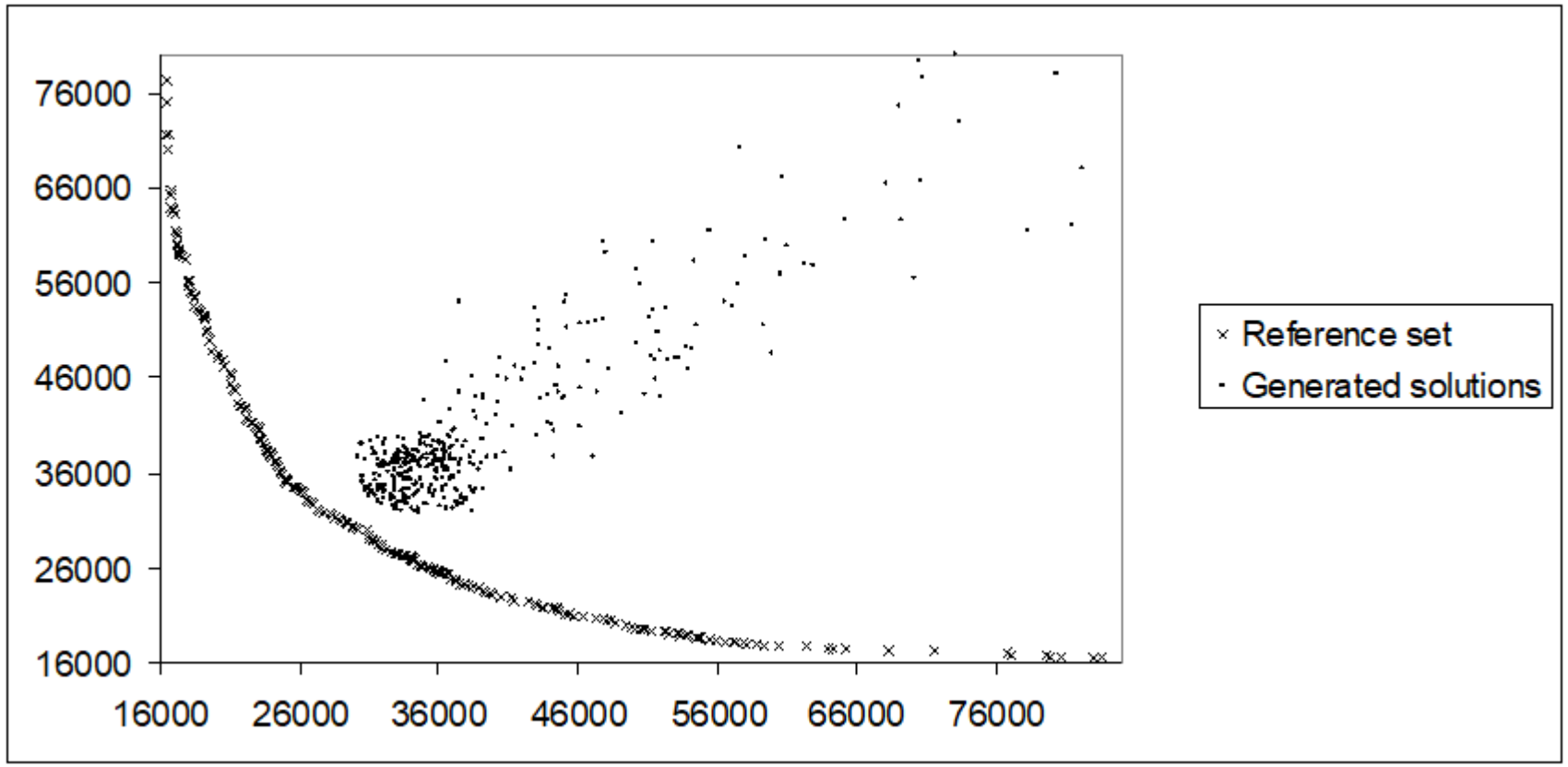


$\lambda_1 < \lambda_2$

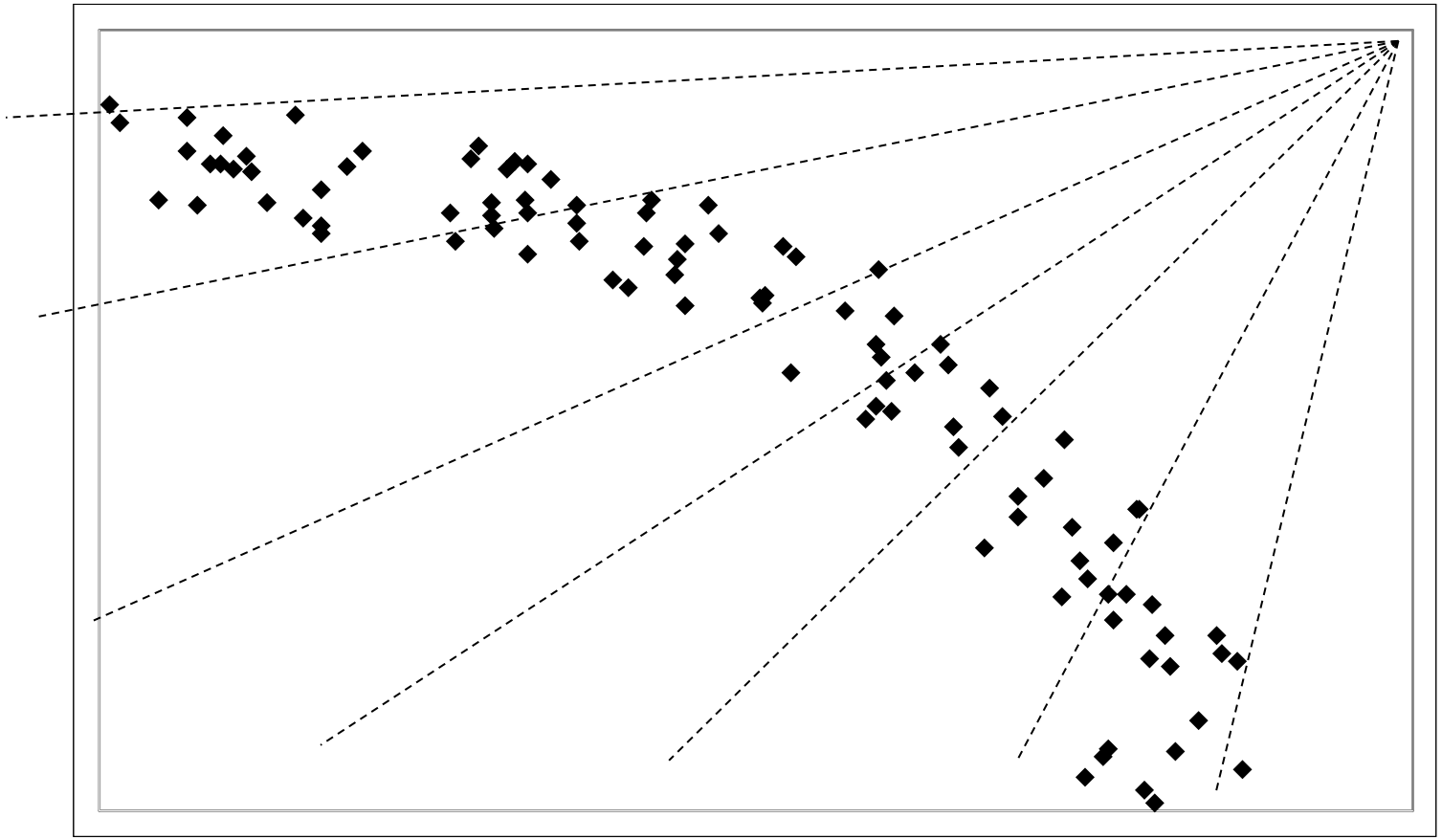




Optimization of a single scalarizing function



Different directions (weight vectors) allow to cover the entire Pareto set/front



Decomposition/scalarization-based algorithms

- Predefined directions (weight vectors)
  - MOSA, MOEA/D
- Random direction selection in each iteration
  - MOGLS
- Common population for all directions – the effect should be better than independent optimization of a certain number of scalarizing functions
  - MOEA/D, MOGLS

Indicator-based algorithms

- Quality indicator – a measure of the quality of the approximation of a Pareto set by a set of solutions (e.g. by the current population)
- New solutions evaluated in terms of the impact/contribution to this indicator
- The most popular indicator – hypervolume of an area dominated by at least one solution
  - Advantage – compatibility with the dominance relationship – adding a new dominant or non-dominant solution will always improve this indicator
  - Problems – high computational complexity – exponential growth as the number of objectivesincreases
- Also widely used to evaluate the results of other multiobjective algorithms

Main own work in this area

- Pareto simulated annealing – one of the first methods based on simulated annealing, the first population-based
  - Pareto simulated annealing—a metaheuristic technique for multiple-objective combinatorial optimization. P Czyżak, A Jaszkiewicz, Journal of Multi-Criteria Decision Analysis 7 (1), 34-47, 1998.
- One of the first papers on the evaluation of sets of solutions that are approximations of Pareto sets
  - Evaluating the quality of approximations to the non-dominated set, MP Hansen, A Jaszkiewicz, IMM, Department of Mathematical Modelling, Technical Universityof Denmark
- One of the first hybrid evolutionary algorithms based on scalarization/decomposition (the first effective)
  - Genetic local search for multi-objective combinatorial optimization, A Jaszkiewicz, European journal of operational research 137 (1), 50-71, 2002.
  - On the performance of multiple-objective genetic local search on the 0/1 knapsack problem-a comparative experiment, A Jaszkiewicz, IEEE Transactions on Evolutionary Computation 6 (4), 402-412, 2022

Main own work in this area

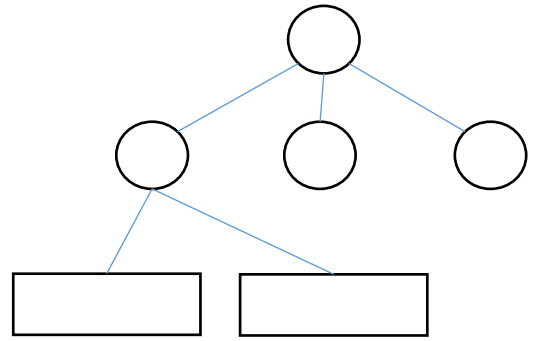
- Pareto Local Search – the best (at the time of publication) results for bi-objective TSP
  - Speed-up techniques for solving large-scale biobjective TSP, T Lust, A Jaszkiewicz, Computers & Operations Research 37 (3), 521-533, 2010
  - Proper balance between search towards and along Pareto front: biobjective TSP case study, A Jaszkiewicz, T Lust, Annals of Operations Research 254 (1), 111-130, 2017
- Effective filtering of potentially Pareto-optimal solutions
  - ND-tree-based update: a fast algorithm for the dynamic nondominance problem, A Jaszkiewicz, T Lust, IEEE Transactions on Evolutionary Computation 22 (5), 778-791, 2018
- Local Pareto search for more than two criteria – selection of promising solutions for neighborhood exploration, use of ND-Tree
  - Many-objective Pareto local search, A Jaszkiewicz, European Journal of Operational Research 271 (3), 1001-1013, 2018
- Analysis of many-objective problems
  - What if we increase the number of objectives? Theoretical and empirical implications for many-objective combinatorial optimization,R Allmendinger, A Jaszkiewicz, A Liefooghe, C Tammer Computers & Operations Research 145, 2002.

Main own work in this area

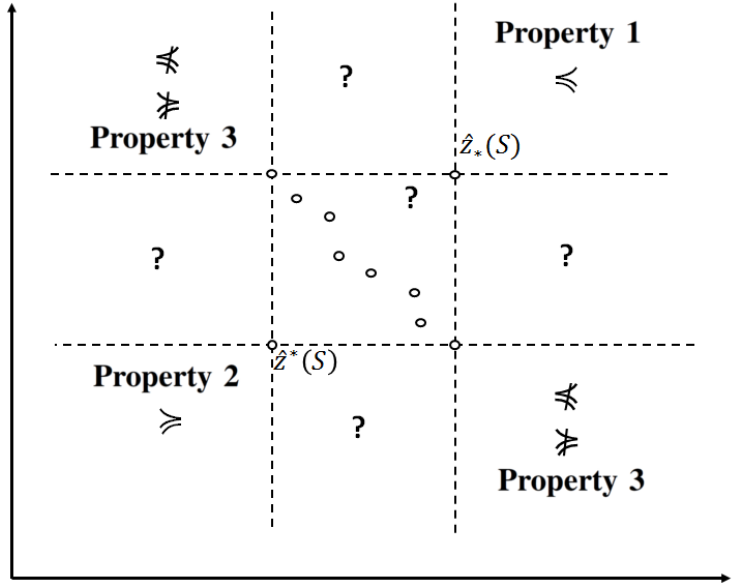
- Hypervolume and R2 calculation algorithms
  - Improved quick hypervolume algorithm, A Jaszkiewicz, Computers & Operations Research 90, 72-83, 2018
  - Approximate Hypervolume Calculation with Guaranteed or Confidence Bounds, A Jaszkiewicz, R Susmaga, P Zielniewicz, International Conference on Parallel Problem Solving from Nature, 215-228, 2020
  - Quick Extreme Hypervolume Contribution Algorithm, A Jaszkiewicz, P Zielniewicz, GECCO 2021, 412–420.
  - Greedy Decremental Quick Hypervolume Subset Selection Algorithms. Andrzej Jaszkiewicz and Piotr Zielniewicz. 2022. PPSN XVII, 164–178.
  - On-line Quick Hypervolume Algorithm. Jaszkiewicz A., Zielniewicz P. GECCO '23: 371-374
  - Andrzej Jaszkiewicz and Piotr Zielniewicz. Improving the efficiency of the distance-based hypervol-ume estimation using ND-Tree. IEEE Transactions on Evolutionary Computation, pages 1–1, 2024.
  - Andrzej Jaszkiewicz and Piotr Zielniewicz. Exact calculation and properties of the R2 multiobjectivequality indicator. IEEE Transactions on Evolutionary Computation, pages 1–1, 2024.

ND-Tree data structure for Pareto archive update

- Recursive division of the Pareto archive into subsets of closely located solutions
- Sub-linear average computational complexity

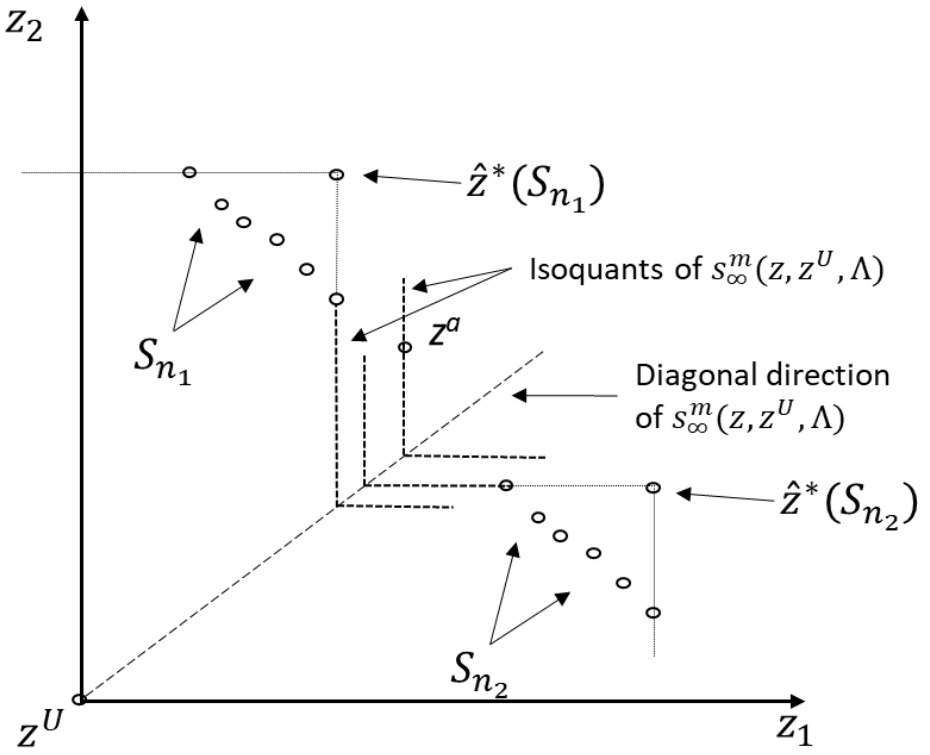


- ND-Tree "to consistently equal or better the run-time performances of the alternatives — on two very different computational architectures — often realising several orders-of-magnitude improvements in run-time„
  - J. E. Fieldsend, “Data structures for non-dominated sets: Implementations and empirical assessment of two decades of advances,” in Proceedings of the 22nd Annual Conference on Genetic and Evolutionary Computation. AMC, 2020, p. 489–497
- A. Jaszkiewicz and T. Lust, “ND-Tree-Based Update: A Fast Algorithm for the Dynamic Nondominance Problem,” IEEE Transactions On Evolutionary Computation, vol. 22, no. 5, pp. 778–791, OCT 2018.



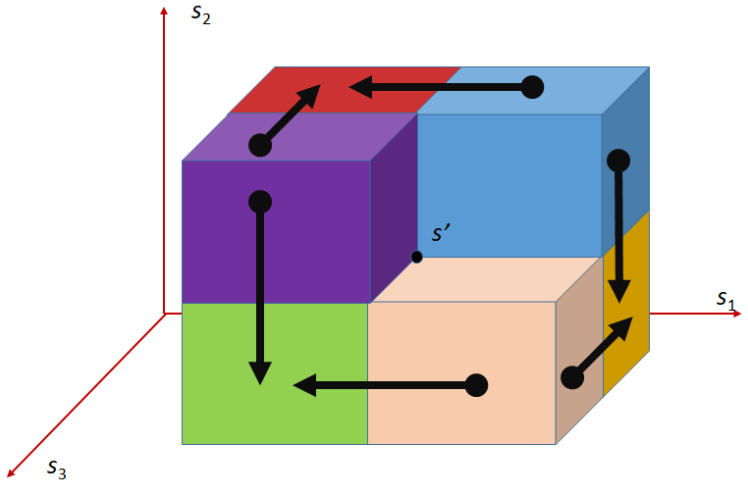
Other applications of ND-Tree

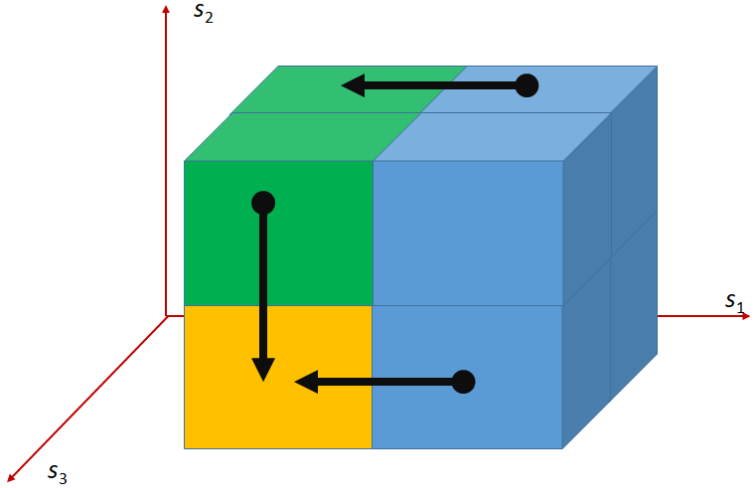
- Hypervolume estimation by Monte Carlo sampling
  - Jaszkiewicz, A., Susmaga, R., and Zielniewicz, P. (2020). Approximate hypervolume calculation with guaranteed or confidence bounds. PPSN 2020, 215–228.
- Finding the minimum/maximum of a (modified) Chebyshev scalarizing function in sublinear time
  - Jaszkiewicz A., Many-Objective Pareto Local Search, European Journal of Operational Research, Volume 271, Issue 3, 2018, Pages 1001-1013
  - Jaszkiewicz, A., Zielniewicz, P. (submitted). ND-Tree-based hypervolume estimation for large solution sets



Improved quick hypervolume algorithm

- Two methods of division into
  - $2^d - 2$ subproblems QHV
  - *d* subproblems improved QHV
- IQHV better computational complexity and practical behavior

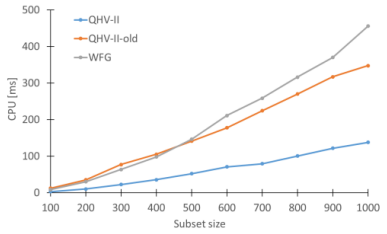




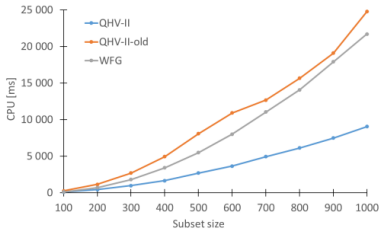
- A. Jaszkiewicz. 2018. Improved quick hypervolume algorithm. Computers and Operations Research (2018), 72–83.
- Andrzej Jaszkiewicz and Piotr Zielniewicz. 2022. Greedy Decremental Quick Hypervolume Subset Selection Algorithms. PPSN XVII, 164–178.

Improved quick hypervolume algorithm

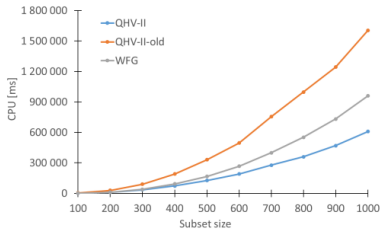
- State-of-the-art for exact calculation of hypervolume



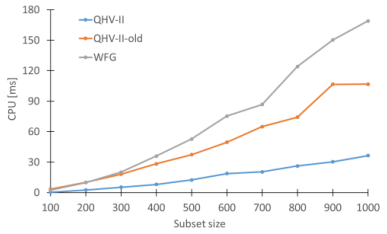
concave, 6



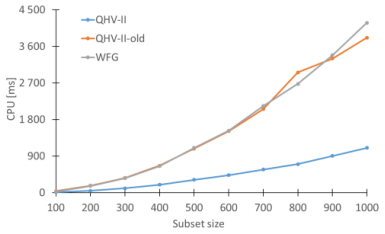
concave, 8



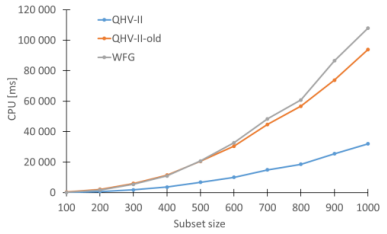
concave, 10



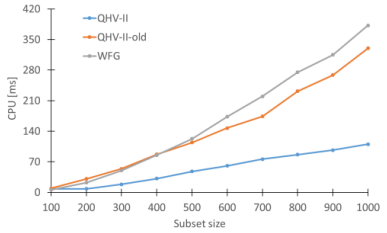
convex, 6



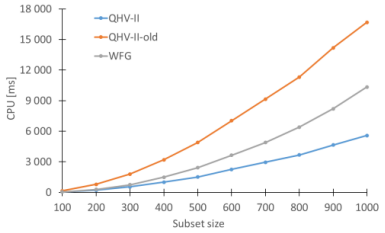
convex, 8



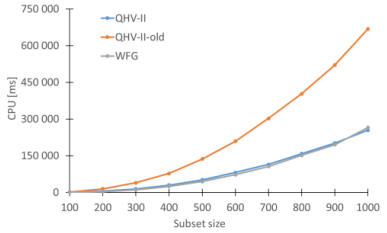
convex, 10



linear, 6



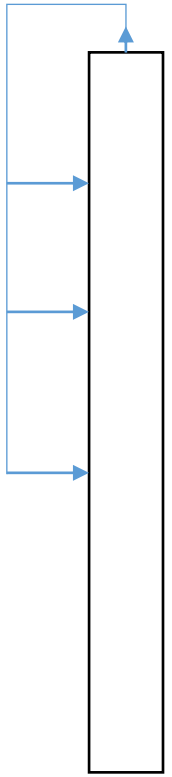
linear, 8



linear, 10

Approximate Hypervolume Calculation with Guaranteed or Confidence Bounds

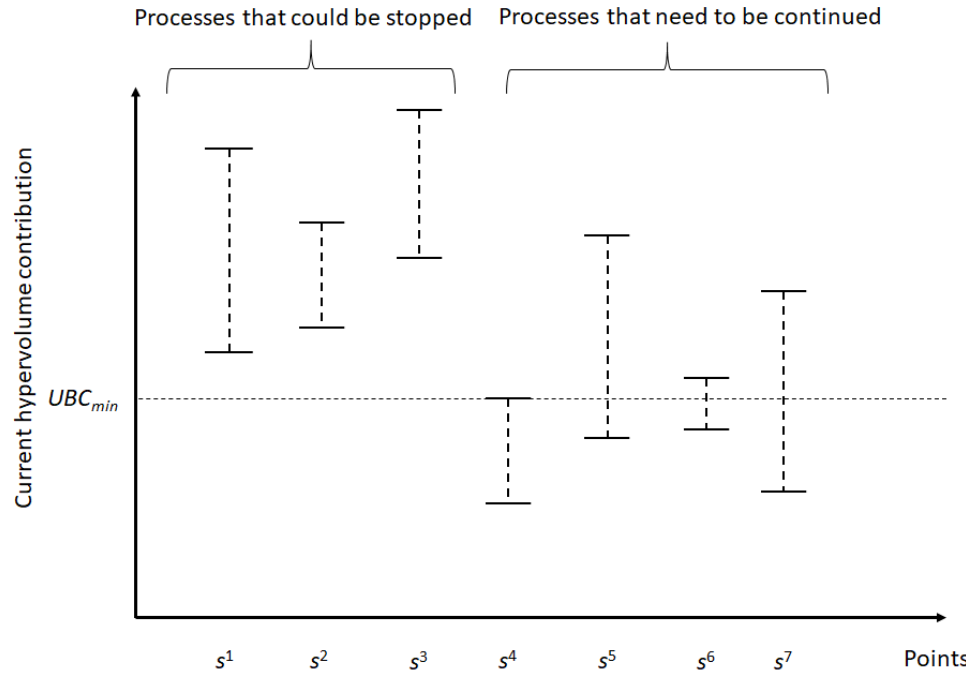
- In each step of IQHV, it is possible to calculate the lower and upper bound (range) of HV
- Each new subproblem is smaller than the source
- Switching from depth first search (IQHV) to search based on subproblem size – priority queue – assures fast convergence of the HV range
- Possible switch to estimation by Monte Carlo sampling – using ND-Tree



- Jaszkiewicz, A., Susmaga, R., and Zielniewicz, P. (2020). Approximate hypervolume calculation with guaranteed or confidence bounds. PPSN 2020, 215–228.

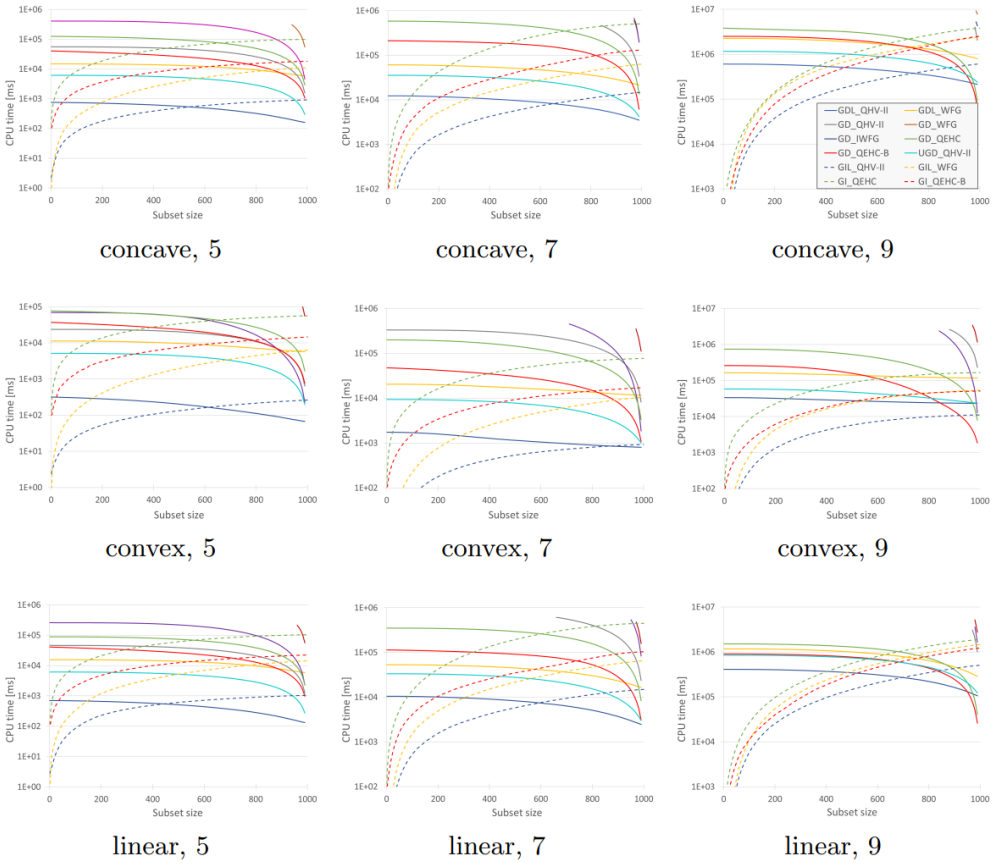
Searching for the solution with the highest/lowest HV contribution

- HV contribution – HV difference with and without this solution – a measure of the contribution of the solution to the quality of the entire set
- Often used in indicator based algorithms
- The calculation of the HV contribution comes down to a smaller problem of calculating the HV
- Idea – parallel calculation of HV contribution for each solution and use of bounds calculated with the previous algorithm
- Jaszkiewicz, A., Zielniewicz, P.: Quick Extreme Hypervolume Contribution Algorithm, p. 412–420, GECCO 2021



Greedy Decremental Quick Hypervolume Subset Selection

- Selection of the subset of solutions with the highest HV
- Greedy decrement algorithm
- Using HV submodularity – greedy lazy approach



concave, 5 | concave, 7 | concave, 9

convex, 5 | convex, 7 | convex, 9

linear, 5 | linear, 7 | linear, 9

- Andrzej Jaszkiewicz and Piotr Zielniewicz. 2022. Greedy Decremental Quick Hypervolume Subset Selection Algorithms. PPSN XVII, 164–178.

On-line Quick Hypervolume Algorithm

- Updating HV by adding HV contribution
- E.g. trajectory of HV evolution during the run of an evolutionary algorithm
  - On-line Quick Hypervolume Algorithm. Jaszkiewicz A., Zielniewicz P. GECCO '23: 371-374

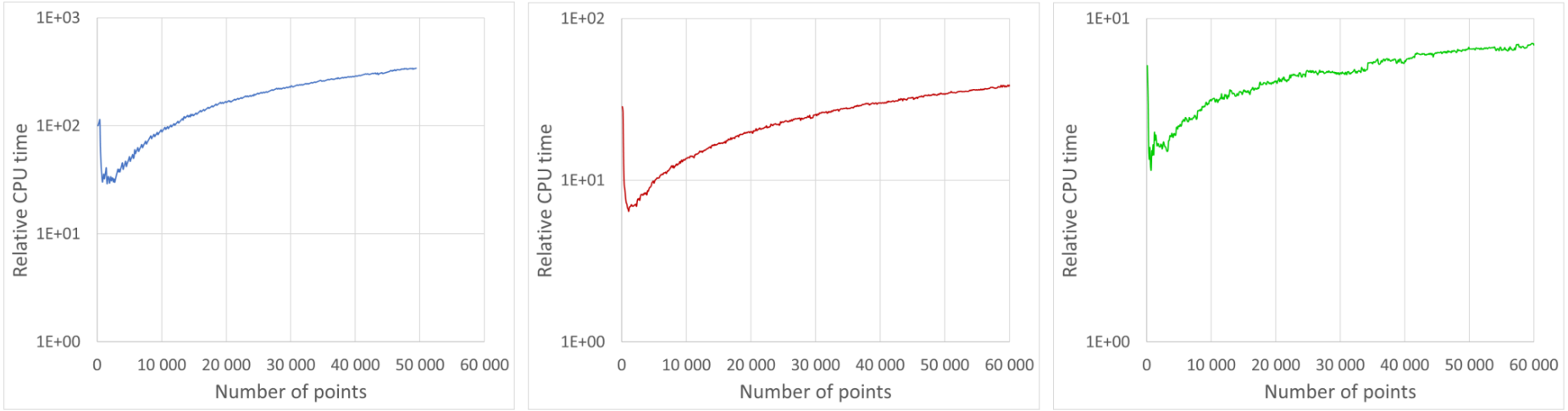


Figure 4: Relative running times of on-line QHV w.r.t. the running time of the static calculation of the hypervolume on MOGLS data sets

Further research plans

- Algorithms based on QR2 analogous to QHV – approximation with lower/upper limit, best/worse solution selection, subset selection, on-line update
- Integration with indicator-based EMO, e.g. SMS-EMOA - both R2 and HV
- Subset selection based on R2 through exact optimization (MP solvers)
- Application of Grover's quantum algorithm in local search
- Applications:
  - Vehicle routing and related problems
  - Selection of satellite images
  - Optimization of computing workloads on advanced architectures
  - Evolutionary simplification of neural networks
  - Cybersecurity, e.g. role mining problem
  - (Generative) neural networks and physics, chemistry, mechanics